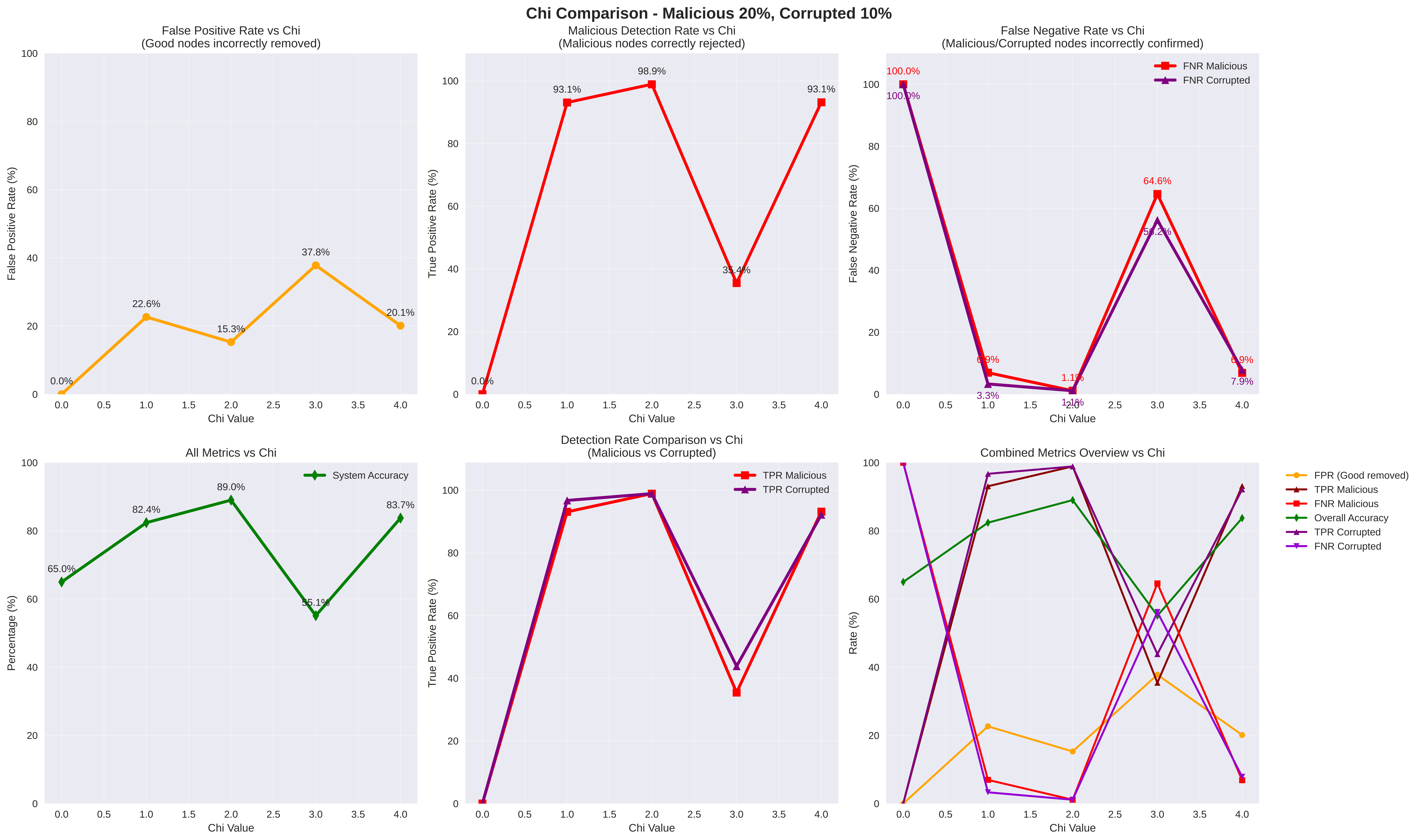
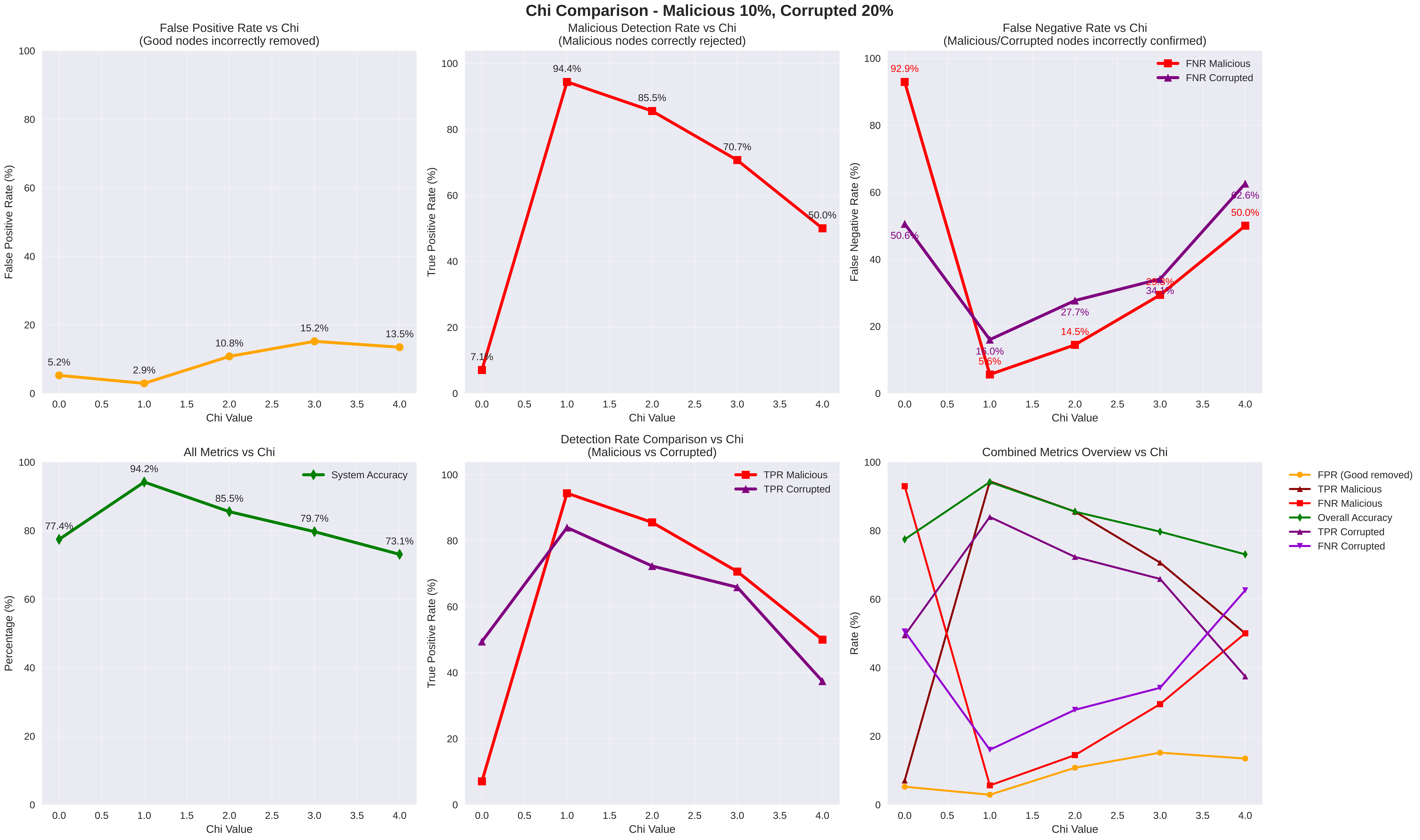
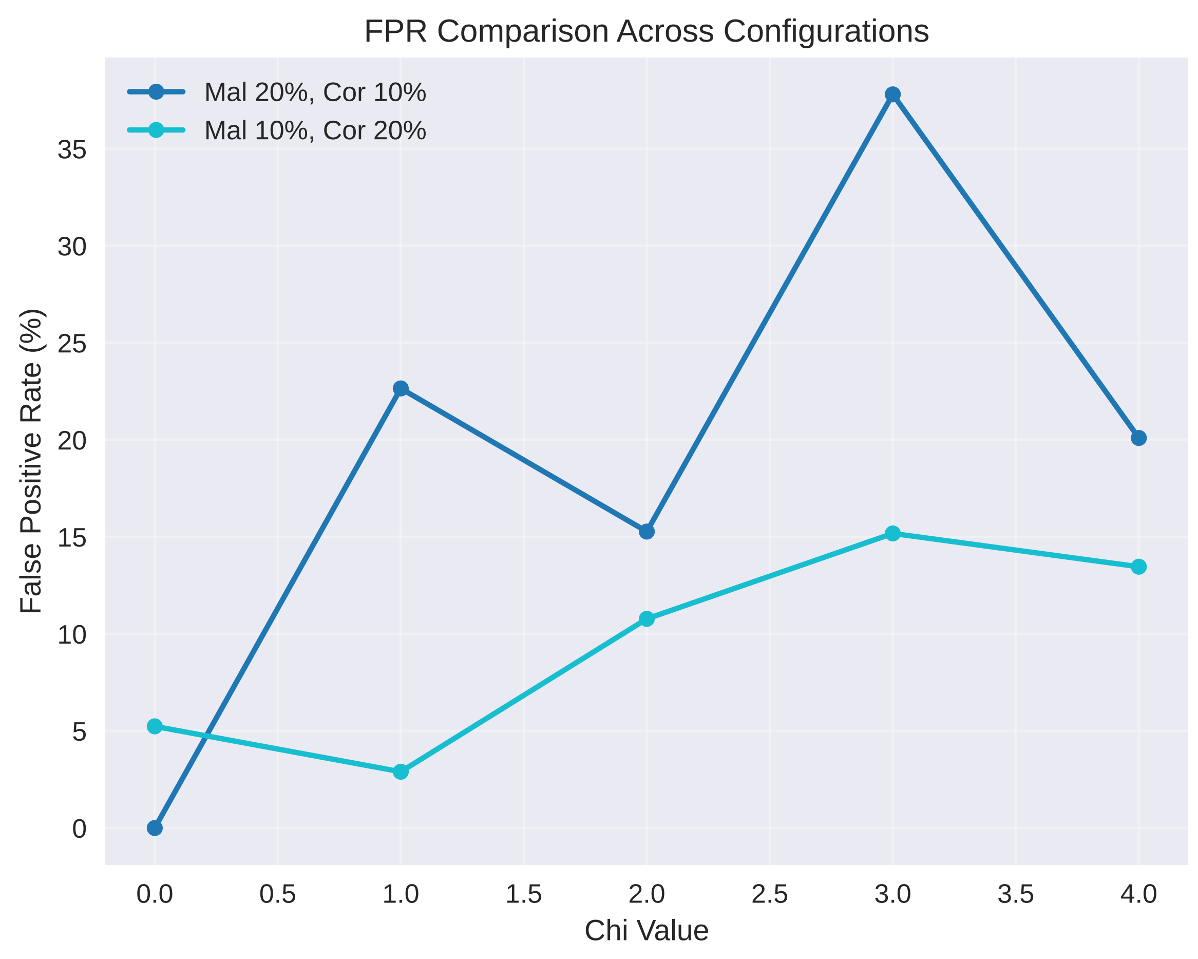
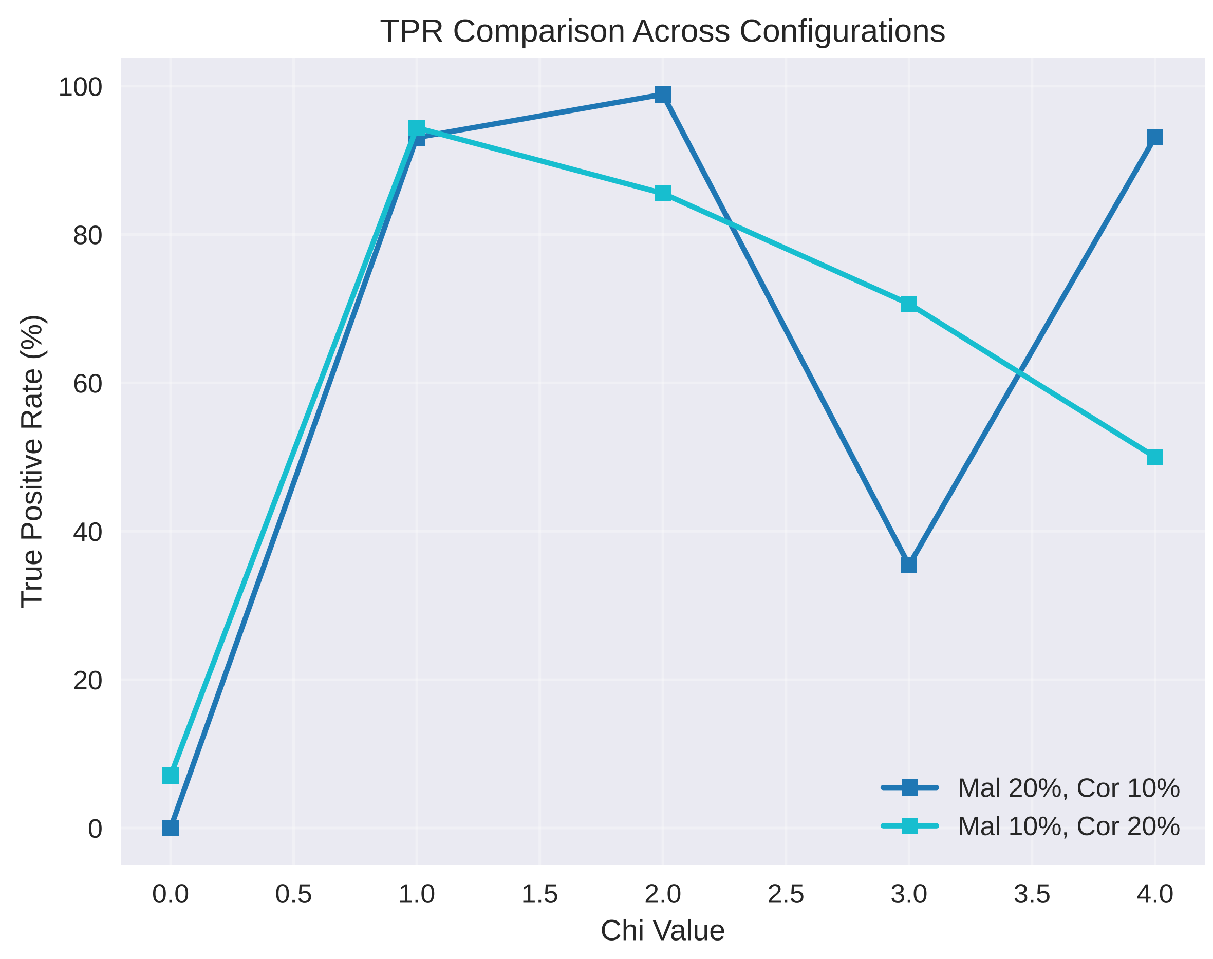

Chi Comparison - Malicious 20%, Corrupted 10%
False Positive Rate vs Chi
(Good nodes incorrectly removed)
False Positive Rate (%)
Chi Value
0.0%
22.6%
15.3%
37.8%
20.1%
Malicious Detection Rate vs Chi
(Malicious nodes correctly rejected)
True Positive Rate (%)
Chi Value
0.0%
93.1%
98.9%
35.4%
93.1%
False Negative Rate vs Chi
(Malicious/Corrupted nodes incorrectly confirmed)
False Negative Rate (%)
Chi Value
FNR Malicious
FNR Corrupted
100.0%
100.0%
6.9%
3.3%
1.1%
1.1%
64.6%
56.2%
6.9%
7.9%
All Metrics vs Chi
Percentage (%)
Chi Value
System Accuracy
65.0%
82.4%
89.0%
55.1%
83.7%
Detection Rate Comparison vs Chi
(Malicious vs Corrupted)
True Positive Rate (%)
Chi Value
TPR Malicious
TPR Corrupted
Combined Metrics Overview vs Chi
Rate (%)
Chi Value
FPR (Good removed)
TPR Malicious
FNR Malicious
Overall Accuracy
TPR Corrupted
FNR Corrupted
0
20
40
60
80
100
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0


Chi Comparison - Malicious 10%, Corrupted 20%
False Positive Rate vs Chi
(Good nodes incorrectly removed)
False Positive Rate (%)
Chi Value
5.2%
2.9%
10.8%
15.2%
13.5%
Malicious Detection Rate vs Chi
(Malicious nodes correctly rejected)
True Positive Rate (%)
Chi Value
7.1%
94.4%
85.5%
70.7%
50.0%
False Negative Rate vs Chi
(Malicious/Corrupted nodes incorrectly confirmed)
False Negative Rate (%)
Chi Value
FNR Malicious
FNR Corrupted
92.9%
50.6%
16.0%
5.6%
14.5%
27.7%
29.3%
34.1%
62.6%
50.0%
All Metrics vs Chi
Percentage (%)
Chi Value
System Accuracy
77.4%
94.2%
85.5%
79.7%
73.1%
Detection Rate Comparison vs Chi
(Malicious vs Corrupted)
True Positive Rate (%)
Chi Value
TPR Malicious
TPR Corrupted
Combined Metrics Overview vs Chi
Rate (%)
Chi Value
FPR (Good removed)
TPR Malicious
FNR Malicious
Overall Accuracy
TPR Corrupted
FNR Corrupted
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
20
40
60
80
100

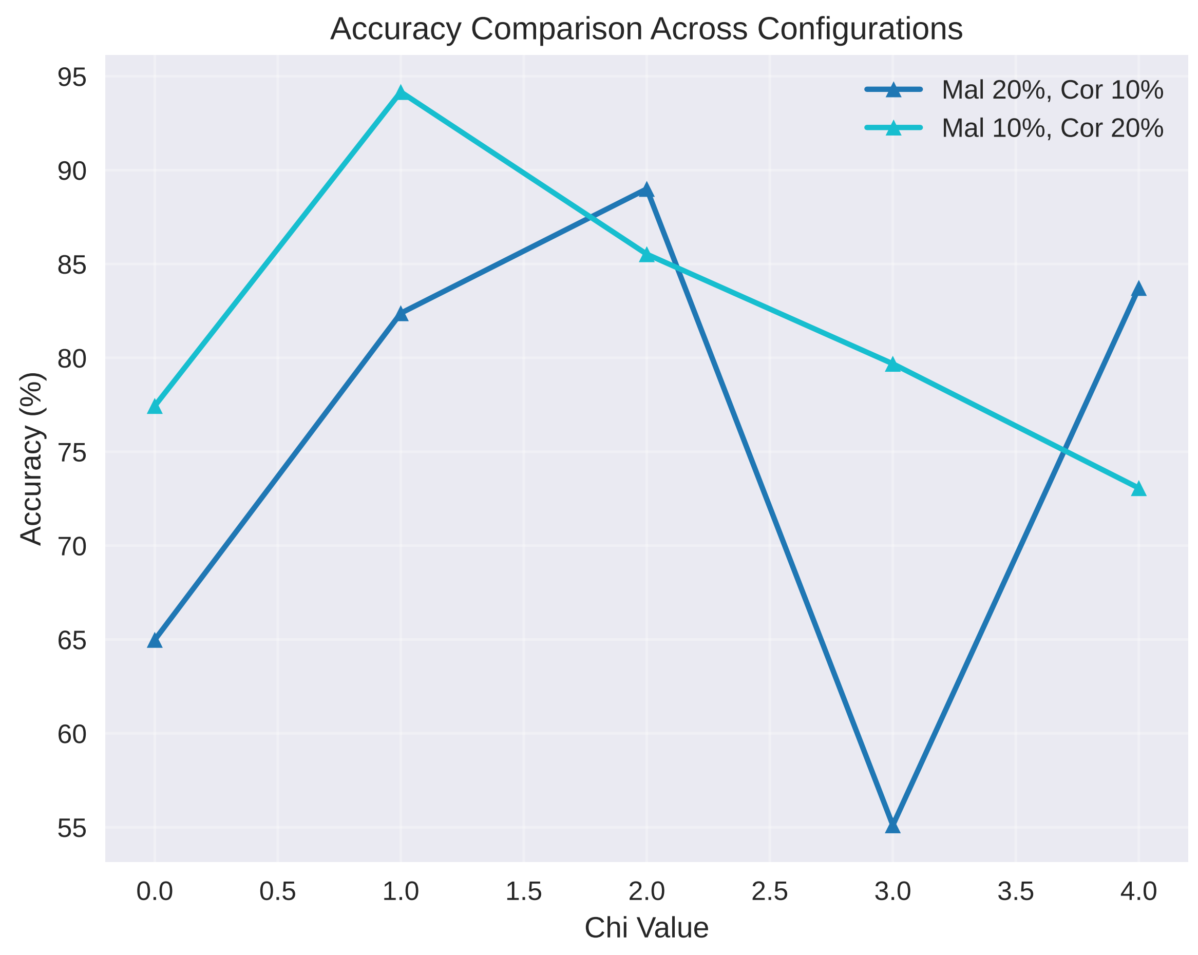
Overall Chi Comparison



FPR Comparison Across Configurations



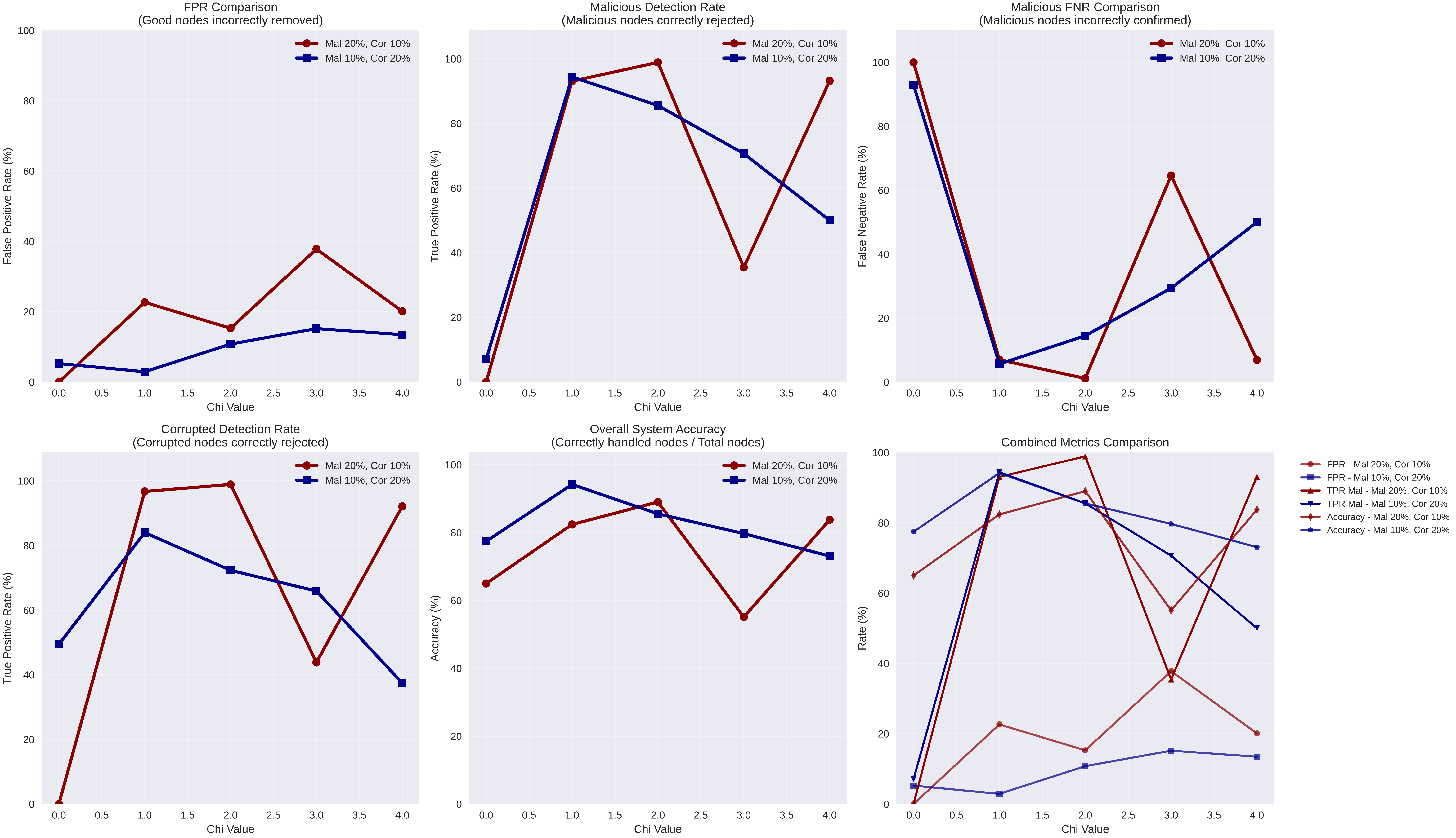
TPR Comparison Across Configurations



Accuracy Comparison Across Configurations

```
CHI EFFECTIVENESS RANKING\n========================================\n\n1. Chi = 1\n   FPR: 12.8%, TPR: 93.7%, FNR: 6.3%, Acc: 88.3%\n   Score: 162.9\n\n2. Chi = 2\n   FPR: 13.0%, TPR: 92.2%, FNR: 7.8%, Acc: 87.3%\n   Score: 158.7\n\n3. Chi = 4\n   FPR: 16.8%, TPR: 71.6%, FNR: 28.4%, Acc: 78.4%\n   Score: 104.8\n\n4. Chi = 3\n   FPR: 26.5%, TPR: 53.0%, FNR: 47.0%, Acc: 67.4%\n   Score: 47.0\n\n5. Chi = 0\n   FPR: 2.6%, TPR: 3.5%, FNR: 96.5%, Acc: 71.2%\n   Score: -24.3\n\n
```

Direct Configuration Comparison: Mal 20%/Cor 10% vs Mal 10%/Cor 20%



Summary Statistics

```
CHI COMPARISON ANALYSIS - SUMMARY
================================

Generated: 2025-08-30 16:44:03
Results Path: FL/results

CONFIGURATIONS ANALYZED:
\n• Malicious 20%, Corrupted 10%\n  Chi values: [0, 1, 2, 3, 4]\n    Chi 0: FPR=0.0%, TPR=0.0%, FNR=100.0%, Acc=65.0%\n    Chi 1: FPR=22.6%, TPR=93.1%, FNR=6.9%, Acc=82.4%\n    Chi 2: FPR=15.3%, TPR=98.9%, FNR=1.1%, Acc=89.0%\n    Chi 3: FPR=37.8%, TPR=35.4%, FNR=64.6%, Acc=55.1%\n    Chi 4: FPR=20.1%, TPR=93.1%, FNR=6.9%, Acc=83.7%\n\n• Malicious 10%, Corrupted 20%\n  Chi values: [0, 1, 2, 3, 4]\n    Chi 0: FPR=5.2%, TPR=7.1%, FNR=92.9%, Acc=77.4%\n    Chi 1: FPR=2.9%, TPR=94.4%, FNR=5.6%, Acc=94.2%\n    Chi 2: FPR=10.8%, TPR=85.5%, FNR=14.5%, Acc=85.5%\n    Chi 3: FPR=15.2%, TPR=70.7%, FNR=29.3%, Acc=79.7%\n    Chi 4: FPR=13.5%, TPR=50.0%, FNR=50.0%, Acc=73.1%\n
\nMETRICS EXPLANATION:
• FPR (False Positive Rate): Good nodes incorrectly removed / Total good nodes sent
• TPR (True Positive Rate): Malicious nodes correctly detected / Total malicious nodes sent
• FNR (False Negative Rate): Malicious nodes incorrectly confirmed / Total malicious nodes sent
• Accuracy: Overall system accuracy (correctly handled nodes / total nodes)

CALCULATION METHOD:
• Each round selects K-1 = 9 nodes randomly from K = 10 peers
• Expected nodes per type = (9 × peer_count_of_type) / 10 × total_rounds
• Sums Good_Nodes_Removed_This_Round and Malicious_Nodes_Confirmed_This_Round across ALL rounds
• Uses selection probability based on peer distribution

OPTIMAL VALUES:
• FPR: 0% (no good nodes removed)
• TPR: 100% (all malicious nodes detected)
• FNR: 0% (no malicious nodes confirmed)
• Accuracy: 100% (perfect classification)

CHI PARAMETER:
The chi parameter controls the depth of past cone inspection
in the TrustInformedTipSelector_NoThreshold algorithm.
Higher chi values inspect deeper history for similarity analysis.
```